

Task 3:- OS interaction and landscape

OS interfaces :-

Now that you have solid understanding of OS and its various responsibilities, let's look at how we interact with OS. Interaction with OS can be divided into two main parts: The Graphical user Interface (GUI) and the Command line Interface (CLI).

Graphical user Interface :-

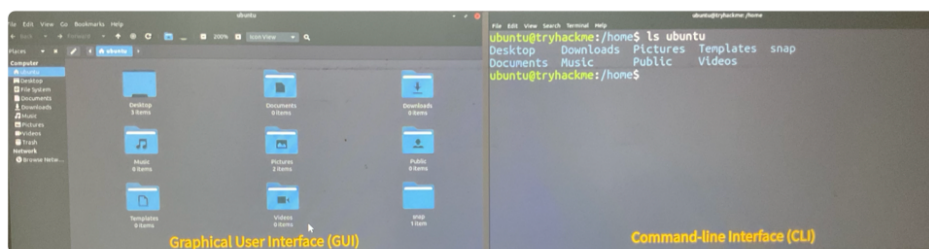
GUI is what you most likely used to interacting with. It provides graphical representation of all information you want to access on your computer. Think of folder icons, windows for your applications, and menus for settings. You tap an icon of place you want to visit, app generates direction for you.

Command line Interface :-

The CLI is where you enter specific text based commands to retrieve or manipulate information. Instead of clicking icons, you tell computer exactly what you want using words and syntax that system understands. This gives you more precision, control and speed. ^{but it requires familiarity with commands.} Later we learn CLI on both linux and windows learn how to navigate files, inspect system

information, and interact with your OS beyond GUI.

In this case below, to display content ^{ubuntu} uses home directory. GUI requires few clicks for folder navigation.



Operating System landscape :-

Now it's time to look at bigger picture. Not all operating system are same. Different devices and jobs demand different designs, ranging from your phone to webserver in data center.

Operating System Type	Primary Use Case	Key Characteristics
Desktop	Personal computers, daily work, gaming, content creation	Rich graphic interface, runs many apps at once, user-focused

Server	Web hosting, databases, cloud services, back-end	Headless (no GUI), maximum uptime, multi-user, remote access
Mobile	Smartphones and tablets	Touch-based UI, power efficient, always connected, app sandboxing
Embedded	Appliances, cars, IoT devices, smart TVs, routers	Tiny footprint, runs on limited hardware
Virtual/Cloud	Lab machines, containers, cloud instances	Lightweight, scalable, rapid deployment

Real world operating Systems: -

Let's look at major families of operating systems you'll encounter in real world. We'll highlight common versions or distributions you'll see in each and follow same category as above:

Desktop: -

* **Windows**: - The most widely used OS on personal computers. Windows 11, Windows 10 (End of life)

* **macOS**: - Apple's desktop OS, known as its polished GUI and integration with other Apple devices. ^{Sonoma (14), Sequoia (15).}

* **Linux**: - Not a single OS but a family of open source operating system called distributions. ^{Ubuntu, Debian, Fedora.}

Server: -

* **Windows**: - used in large networks, data centers, and corporate environment. ^{Server 2016, 2019, 22, 2025.}

* **Linux**: - Vast majority of web servers, trusted for its reliability and open source nature. ^{Ubuntu, Debian, Red Hat.}

* **Unix**: - larger enterprises, finance, telecom, government. ^{IBM AIX, Oracle Solaris.}

Mobile: -

* **Android**: - The most widely used mobile OS, which runs on phones, tablets and smart devices. ^{Android 14-16, Manufacturer view.}

* **iOS**: - Apple's mobile OS running on iPhones, iPads and other devices. ^{iOS 17, 18, 26.}

Embedded and IOT Devices: -

* **Embedded Linux**: - Specialized OS built into devices with dedicated functions.

^{OpenWRT, Ubuntu Core, Yocto project.}

* **Real Time OS**: - Designed for apps where tasks need guaranteed response times

^{FreeRTOS, VxWorks, QNX}

Virtual and cloud: -

* **Cloud IaaS**: - Massive data centers that host websites, apps and streaming services. ^{Ubuntu LTS, Amazon Linux, Rocky Linux}

* **Containers - Optimized**: - Lightweight alternatives to VMs that reduce cost and speed up deployment.

dependencies Alpine Linux, Bottlerocket AWS, Flatcar Linux.

Why so many operating systems?

Different devices and environments require different capabilities from OS. A laptop must be user friendly and support multitasking. Servers require stability, security and must be able to run continuously without interruption. Mobile devices need power efficiency and hardware integration to extend battery life. Embedded system uses lightweight operating systems designed for special purpose.

The companies and communities develop their OS also have their own goals. Some focus on security, openness or customization. Because each environment values different capabilities.

Key Terminologies:-

OS:- The core software that manages hardware, applications and all system resource.

kernel space:- The OS highly privileged area with direct hardware access, home of kernel, which directly manages hardware and system resource.

user space:- The area which regular applications run with limited permission for safety and system stability.

(GUI)
Graphical user interface:- The visual part of OS, windows icon, menus that let you interact through clicking and tapping.

Command line interface (CLI):- A text based interface, where you type control system with speed and precision.